DST NM-ICPS Technology Innovation Hub on Autonomous Navigation and Data Acquisition Systems (UAVs, ROVs, etc.) – TiHAN at IIT Hyderabad

Annexure A: Proforma for submission of Project Proposal

OVERVIEW

1. Project Title

Design and Development of Efficient Road Object Detection Methods in Extreme Weather Conditions for Secure Autonomous Driving

2. Organization

a) *Indian Institute of Technology Tirupati.*

b) *Chindepalle, Andhra Pradesh - 517619.*

c) *Academic Institution (Public Technical Institute)*

3. Management

PI*: Dr. Chalavadi Vishnu,* ***Co-PI****: Dr. Yeturu Kalidas*

4. Duration of the Project

8 months

Dr. Chalavadi Vishnu, Assistant Professor, Computer Science and Engineering, Indian Institute of Technology Tirupati

5. Context/Background

Advanced Driver Assistance Systems (ADAS) in autonomous driving have become a cornerstone of modern transportation research, offering promises of enhanced road safety, efficiency, and reduced human dependency. A critical enabler of this technology is the ability of ADAS to perceive, interpret, and act upon their surroundings with precision and reliability. However, achieving this goal requires overcoming significant challenges in adverse weather, specifically extreme weather scenarios that can obscure vision, reduce sensor reliability, and complicate decision-making processes.

This project addresses these challenges by proposing solutions to the following tasks:

Extreme Weather Understanding *- Generally, vision and LIDAR sensors in advanced driver-assisted systems (ADAS) are modeled from extreme weather data (fog, snow, rain, sandstorm), which is quite scarce - DAWN dataset, for example, often requiring considerably more data for the system to understand the weather in the environment better.*

Road Object Detection in Extreme Weather *- Following along the extreme weather, as shown in Fig. 1, the objects on the road and their visibility also affects the ADAS sensors, lane detection, and path planning systems by abrupt weather changes.*

Fig.1 - Detection of objects in extreme weather

Computationally Intensive *- Whenever the environment changes, the ADAS sensors have to utilize more resources due to the selection of deep learning architectures, which might help in understanding the extreme weather along with objects in it, but, will require additional computation, memory, and, in turn, fuel/battery power.*

Prior research has placed limited emphasis on understanding such extreme weather conditions along with identifying objects, in real-time in an efficient way from an autonomous perspective. There are numerous applications of extreme weather understanding in ADAS, like scene perception and path planning. Identifying the background weather of an image can help to choose the lightweight and heavyweight deep learning models appropriately to reduce the computational overhead and ensure efficient management of resources. Deep learning models like VGG16, Resnet, and Faster R-CNN are commonly utilized for perception tasks like object detection. Typically, these models exhibit superior performance in clear weather conditions. However, they may struggle to perform optimally in such adverse weather conditions where visibility is significantly reduced. In these cases, heavyweight models, which often require more computational resources, are commonly employed.

Limited data availability is a significant challenge when training deep neural networks in inclined weather conditions. Perception tasks like scene classification and traffic detection require huge training data. Failure to train the model with enough data leads to overfitting. There are several publicly available driving datasets, like Dawn Dataset, Cityscapes, and BDD100K. Unfortunately, most of these datasets primarily consist of images captured in clear weather conditions. As a result, there is a scarcity of datasets that specifically encompass images captured in extreme weather conditions. Extreme weather data unavailability is a major bottleneck due to cost, time, and difficulty in capturing. However, there has been considerable progress in synthetic weather generation for AV technology [1],

Therefore, our work primarily emphasizes weather transition data generation using generative approaches. Further, our work focuses on classifying transition states to enhance scene perception and effectively utilize computational resources. Later, we introduce a domain adaptive object detection mechanism to identify road elements in unseen transitional weather conditions.

References

[1] T. Rothmeier and W. Huber, “Let it snow: On the synthesis of adverse weather image data,” in 2021 IEEE International Intelligent Transportation Systems Conference (ITSC), 2021, pp. 3300–3306.

6. Problems to be addressed

Current ADAS systems excel in clear weather scenarios, but their performance degrades significantly in extreme weather (e.g., rain, fog, or snow). These conditions pose unique challenges, such as rapid variations in visibility due to transitions like fog lifting or rain starting can confuse vision-based perception systems, and can create glare and reflections that obscure objects.

Most existing perception models lack the robustness to handle these conditions, leaving ADAS vulnerable to accidents and system failures. Addressing these scenarios is essential for improving ADAS reliability and safety. These extreme weather conditions introduce subtle challenges, such as intermittent rain or fog, making it difficult to distinguish objects or road boundaries. This project highlights the importance of transitional weather conditions and seeks to develop models that can handle these nuanced scenarios effectively.

Existing object detection models have achieved significant performance in normal scenarios when trained on large-scale, high-quality labeled data. However, deploying these models in extreme weather is challenging due to the domain gap caused by varying extreme weather conditions. Also, these models rely heavily on rich labeled data. Annotating massive amounts of data can be time-consuming and labor-intensive [1].

There are some architectures we can utilize right now, such as transformers, foundation models, etc., that can extract very rich contextual features even under low visibility conditions. Generally, these algorithms are computationally intensive, especially, attention mechanisms, that better understand the contextual features of road objects in various extreme weather conditions would not only lead to perception systems that will be robust against extreme weather. Generally, these algorithms are computationally intensive, but, using techniques such as quantization can massively reduce excessive memory/power usage to detect objects in an efficient/optimized manner.

This also motivates research on unsupervised domain adaptation (UDA), because these systems are prone to adversarial attacks either through nature or conscious malicious actors, so safeguarding them through privacy-preserving adversarial defense techniques against data poisoning attacks of one labeled data from one domain to unlabeled data from other domains.

References

[1] H.-Z. Hao, Y. Cheng, Y. Ji, Y. Li, and C.-P. Liu, "Multi-enhanced adaptive attention network for rgb-t salient object detection," in Proceedings of the International Joint Conference on Neural Networks (IJCNN). IEEE, 2022, pp. 01–08

[2] J. Zhang, K. Yang, C. Ma, S. Reiß, K. Peng, and R. Stiefelhagen, "Bending reality: Distortion-aware transformers for adapting to panoramic semantic segmentation," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 16 917–16 927.

7. Aims and Objectives

Provide clear objectives/goals of the project (Max. 1000 Words).

- *Proposing architectures using attention mechanisms, especially Transformers and Foundation models, to better understand the contextual features in different weather conditions will lead to perception systems that will be robust against extreme weather scenarios.*

- *Using Generative Adversarial Networks (GANs), we will be able to do de-raining, de-foggy, and de-storming in real-time on such images/videos.*

- *Currently, transformers and foundation models require a decent amount of training data, along with using up a lot of GPU memory to train such models. But, when combined with Differential convolutional neural networks, Quantization, and Continual learning, we will be able to significantly reduce the computational complexity of these architectures.*

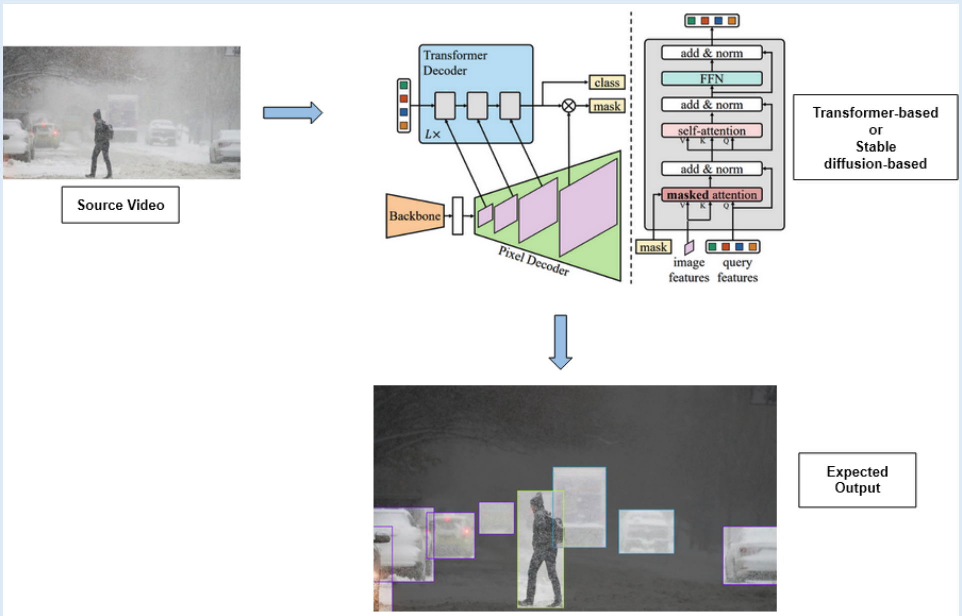
- *Although deep learning architectures are robust against regular noise in the data, these extreme weather conditions, along with malicious actors, will lead to adversarial attacks that fool the autonomous systems into either mis-classifying the road objects or not detecting them at all.*

- *We will develop architectures to generate these adversarial samples to understand these attacks better using GANs and Stable Diffusion models.*

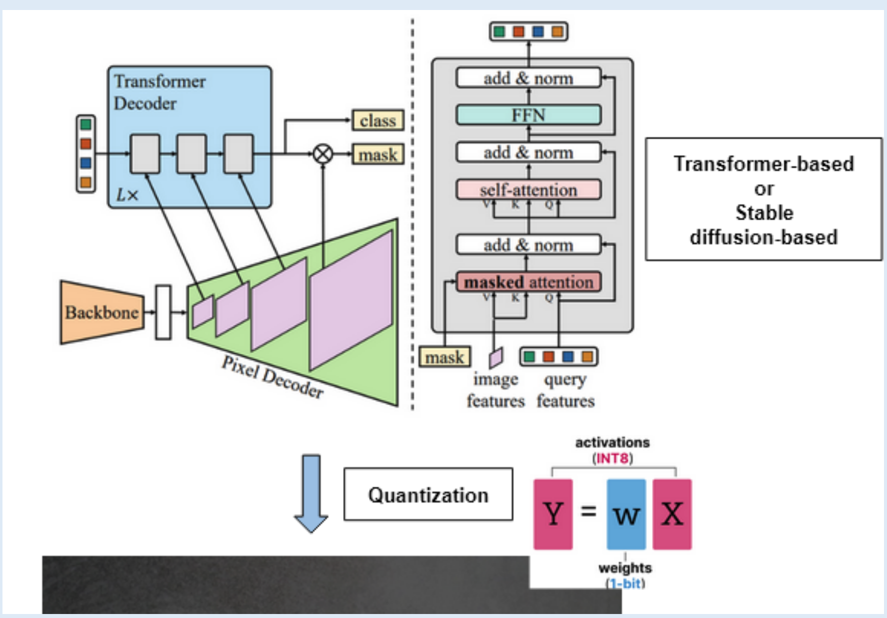
- *We will also propose novel defense algorithms against such attacks using Adversarial training and Defensive distillation techniques as a reference.*

8. Technology

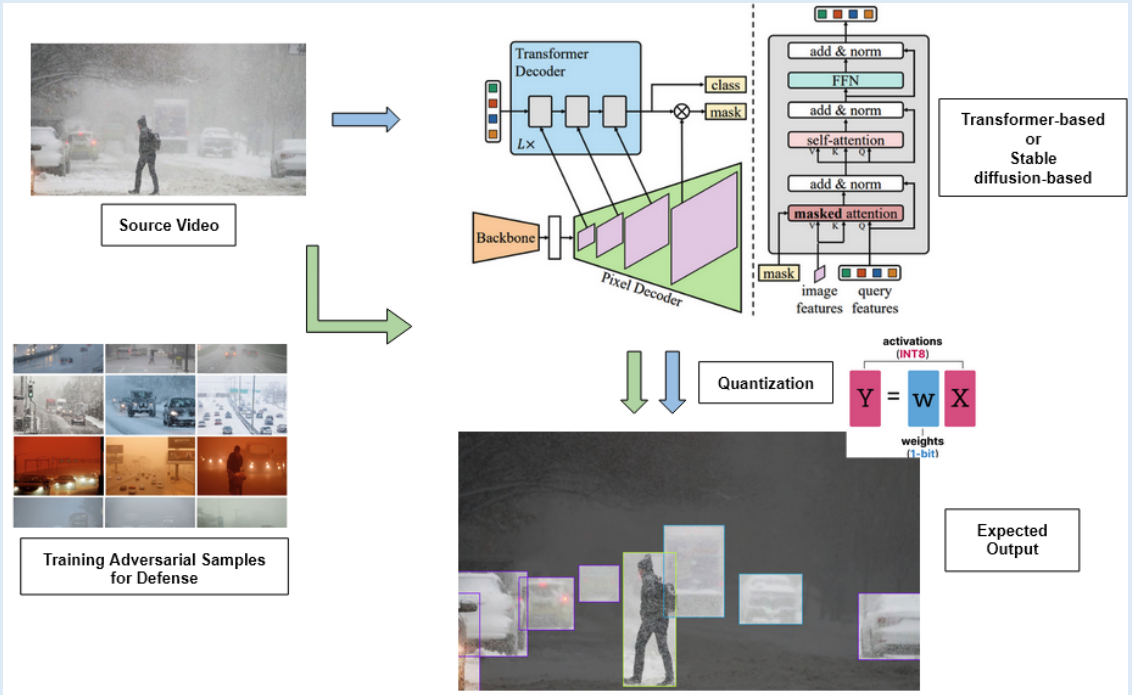
Solution 1: Real-time Road Object Detection under Extreme Weather Conditions using Transformer-based and Stable diffusion-based Architectures.



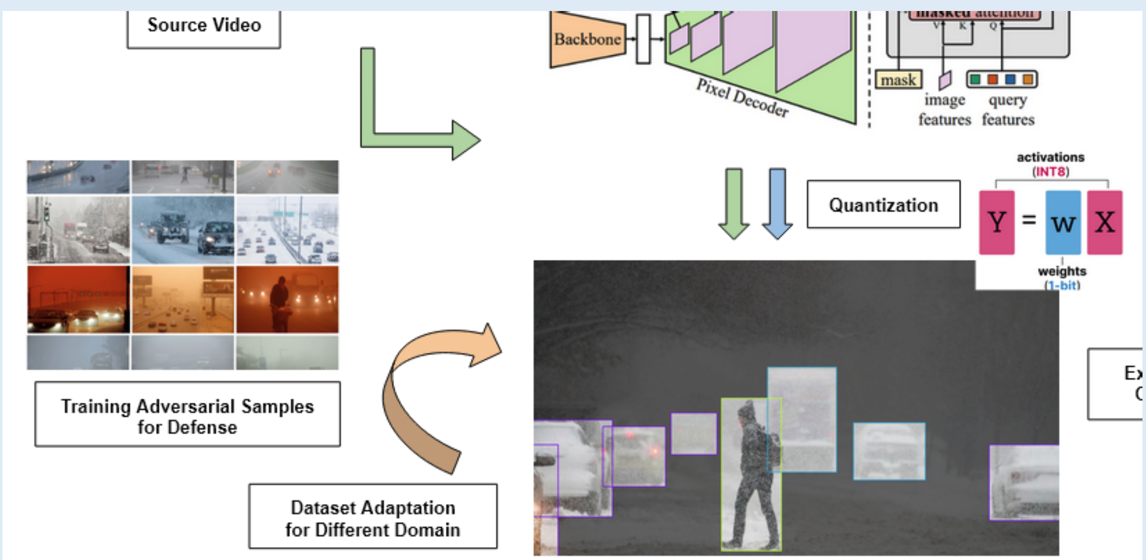
Solution 2: Implementing such a System on Edge Devices and Low Computation Devices of an Autonomous System using Quantization.



Solution 3: Safeguarding against such noise, be it malicious or not, leading to Adversarial Attacks using Defensive Distillation and Adversarial Training.



Solution 4: Robustifying this System against other extreme Conditions for Vulnerable Road Users in Various Other Domains with Domain Adaptation.



9. Timelines

8-month break-up of physical achievements with specific intermediate milestones (in terms of aims and objectives). Provide timelines using the PERT/Gantt chart.

10. Expected Outcomes

Please provide the following:

Specifications of subsystem/system and translational research outcomes

An advanced driver assistance system for secure autonomous driving in extreme weather conditions using efficient object detection techniques

Specifications:

Real-time Performance *- Low latency (<100ms), High frame rate (30+ FPS), Efficient Computation*

Accuracy and Robustness *- Accurate object detection (high precision and recall), Robustness to varying conditions (lighting, weather, occlusions), Reliable operation over extended periods*

Safety and Integration *- Sensor fusion (lidar, radar, etc.), Fail-safe mechanisms (redundancy, fallback strategies), Compliance with safety standards (e.g., ISO 26262)*

Standards:

IEEE P2020 Automotive Imaging *– This was established in order to address the considerable ambiguity in the measurement of image quality of automotive imaging systems, both human and computer vision-based.*

IEEE 2804 *- This standard deals with the performance requirements and test procedures for automotive camera systems.*

IEEE/SAE J3061 *- This standard focuses on cybersecurity for ADAS*

IEEE 802.11P *- This standard defines wireless communication for vehicular environments (WAVE). It enables vehicle-to-everything (V2X) communication, allowing vehicles to exchange data with each other (V2V), infrastructure (V2I), and pedestrians (V2P).*

IEEE 1609 - *This family of standards provides a framework for Wireless Access in Vehicular Environments (WAVE). It covers various aspects of V2X communication*

Technology Readiness Levels:

Expected TRL:2-3 (2 Months)

TRL 2 (Technology concept formulated) *- After 2 months, project planning should be completed and the definition phase. We should have a clear concept of our ADAS system, its functionalities, and target hardware/software. We might have some initial simulations or paper studies.*

TRL 3 (Experimental proof of concept) *– We should develop some basic proof-of-concept implementations or prototypes for core features; we might reach TRL 3.*

Expected TRL:4 (4 Months)

TRL 4 (Technology validated in lab) *- By this point, we should be completing the preliminary design phase. And have validated key technologies in a lab setting. This might involve testing basic algorithms, sensor integration, or software components in a controlled environment.*

Expected TRL:5-6 (6 Months)

TRL 5 (Technology validated in relevant environment) *- After completing detailed design and implementation, we should be validating the integrated system in a more relevant environment.*

TRL 6 (Technology demonstrated in relevant environment) *- Once we have a working prototype that can be demonstrated in a simulated or partially real-world environment (e.g., closed test track), we might reach TRL 6.*

Expected TRL:7-8 (8 Months)

TRL 7 (System prototype demonstration in operational environment) *- By the end of 8 months, we should have a functional prototype that has been tested in an operational environment (on-road testing).*

TRL 8 (System complete and qualified) *- If the system has undergone rigorous testing, optimization, and meets the defined performance and safety requirements, it could reach TRL 8.*

Translational Research Outcomes:

Effective Collision Avoidance System *- Development of more sophisticated and reliable collision avoidance systems.*

Enhanced lane-keeping assist (LKA): *LKA systems are becoming more accurate and reliable, providing smoother lane centering and better assistance in challenging road geometries.*

Skill Development / Workshops (Either of Below):

Details of Workshops, Hackathons, etc.,

Application Workshops:

ADAS in Action: Real-World Applications and Case Studies

The Role of AI and Machine Learning in Advanced Driver Assistance

ADAS for Commercial Vehicles: Enhancing Safety and Efficiency

Hands-on Workshops:

Building an Object Detection System for ADAS (using TensorFlow/PyTorch)

Implementing Sensor Fusion with Lidar and Camera Data

Developing an ADAS Algorithm for Lane Keeping Assist

Simulating and Testing ADAS Systems in a Virtual Environment

Target Audience-Specific Workshops:

ADAS for Automotive Engineers

ADAS for Software Developers

ADAS for Researchers and Academics

Manpower Trained (Level: 4 and No: 2, Either of below)

Computer Vision/AI engineer

Data collector (to record and generate various weather conditions)

Hardware engineer (for sensor integration)

Testing engineer (for testing, data annotation, and system integration)

Plans for Commercialization:

N/A

No. of Technology Products, Publications, IPRs, etc.

2 Publications

11. Target Beneficiaries

The proposed project aims to develop an efficient road object detection system for autonomous vehicles in extreme weather conditions through domain adaptation, attention, generative adversarial networks, etc., The target beneficiaries span a wide range of public and private stakeholders involved in autonomous driving, transportation, and artificial intelligence research. Below is a detailed list of these beneficiaries, along with their roles and potential contributions to the project.

Beneficiaries:

India

Automotive Manufacturers *- Companies like Tata Motors, Mahindra & Mahindra, and Maruti Suzuki are actively integrating ADAS features into their vehicles.*

Tier 1 Suppliers *- Major automotive component suppliers like Bosch, Continental, and WABCO are developing and supplying ADAS technologies (sensors, cameras, ECUs) to vehicle manufacturers.*

Technology Companies *- Infosys, TCS, and KPIT Technologies are providing software development, testing, and validation services for ADAS systems.*

Research Institutions *- IITs and the Centre for Artificial Intelligence and Robotics (CAIR) are conducting research on computer vision, sensor fusion, and ADS technologies.*

Government Initiatives *- The Indian government is promoting road safety and encouraging the adoption of ADAS technologies through initiatives like the Bharat New Vehicle Safety Assessment Program (BNVSAP).*

Abroad

Automotive OEMs (Original Equipment Manufacturers) *- Global automotive giants like Tesla, Mercedes-Benz, BMW, Audi, Volvo, and General Motors are leading the way in ADAS development and deployment, along with some tech giants like Waymo*

Sensor Manufacturers *- Companies like Velodyne Lidar, Luminar, and Mobileye are developing and supplying advanced sensors (lidar, radar, cameras) that are crucial for ADAS.*

Software Companies *- Specialized software companies like Aptiv and Aurora Innovation are developing the software stack and algorithms for autonomous driving and ADAS.*

Relevance

Enhanced Safety *– Warning drivers of potential dangers (e.g., lane departure, blind spot monitoring). Automatically intervening to avoid or mitigate collisions (e.g., automatic emergency braking, collision avoidance). Providing guidance and assistance in extreme weather situations (e.g., lane keeping, adaptive cruise control).*

Enhanced Situational Awareness *- ADAS provides drivers with a better understanding of their surroundings through features like blind spot monitoring, surround-view cameras, and cross-traffic alerts.*

Scalability *- Lightweight models reduce hardware costs, enabling broader market penetration.*

12. Strategy

Please provide the execution strategy for achieving the developmental objectives in the proposed timeline. Also, exploration plans for additional funding through alternate sources (public or private) should also be listed (Max. 500 Words).

The execution strategy for achieving the developmental objectives over 8 months is structured into distinct phases to ensure systematic progress. The initial phase of resource allocation, planning, and design ***(3 months)*** *will focus on the literature review and problem definition to identify gaps in existing classification and detection models for ADAS and finalize technical specifications. This will be followed by the development of novel architectures, annotated data, implementation, and integration testing* ***(3 months)****, where diverse driving scene data for road objects under extreme conditions will be tested and improved along with model optimization. The final stage* ***(2 months)*** *is comprised of a demonstration of the prototype on edge devices, technology transfer, and, if possible, any workshops to bring more awareness to vision security in the autonomous domain.*

13. Legal Framework

Should present the legal framework for the implementation of the project

(Max. 200 Words).

N/A

14. Environmental Impact

State the environmental impact of the proposed objectives (Max. 200 Words)

N/A

15. Finance

Provide half yearly budget with detail breakups including:

Provide list equipment and the quotations of the equipment as supporting document.

Autonomous Vehicle Kit (2) - 2L

*- **PiCar/RoboCar** - 30K INR (each)*

*- **Jetson or RPi** - 20K (each)*

*- **Camera/LiDAR Sensors** - 30K (each)*

*- **Edge TPU** - 20K (each)*

S No	Budget Head	Amount 1st 3 months (₹ Lakhs)	Amount 2nd 3 months (₹ Lakhs)	Amount 3rd 2 months (₹ Lakhs)	Amount Total (₹ Lakhs)
1	Capital Equipment	2,00,000			2,00,000
2	Consumables			40,000	40,000
3	Manpower	1,33,333	1,33,333	1,33,334	4,00,000
4	Travel & Training		50,000	50,000	1,00,000

5	*Technology Development*				
6	*Contingencies / other expenditure*		*7,500*	*7,500*	*15,000*
7	*HRD & Skill Development*	*45,476*			*45,476*
8	*Overheads (5%, no capital)*	*8,940*	*9,541*	*11,542*	*30,023*
	Grand Total (₹ Lakhs)	***3,87,749***	***2,00,374***	***2,42,376***	***8,30,499***

16. Details of Academic Collaborators

Provide list of national or international collaborators. Attach the express of interest from the partners at the end of this proposal.

N/A

17. Details of Industry Partners

Provide list of national or international industry partners. Attach the express of interest from the partners at the end of this proposal.

N/A

Signature of Chief Investigator

C. Vishnu

Designation: Assistant Professor

Date: 06/03/2025

Signature of Head of the Institution/Organization

Designation:

Date: